CSCI313/ECEN428: Software Engineering

Software Requirements Specification

(SRS DOCUMENT)

for

<Airline Reservation System>

Version 1.0

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Mobile Application

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1. Introduction

1.1 Purpose

This document provides a comprehensive overview of the Android application "Skies", which is used to make airline reservations. The system's overall description, interfaces, functional and non-functional requirements, and system diagrams will all be all covered. The previous parts will give a solid understanding of the functions carried out by the system, the usage scenarios, and the constraints that must be satisfied for the system to operate.

1.2 Scope of the project

The Android application “Skies” is an Airline reservations system which works as a mediator between the client and the airline's office to make reservations for air travel without requiring the client to visit the airline's office, saving the client time. The system has one types of users: regular users (Clients) who have registered with the system. After registering with system, the user can login and be allowed to Book a flight, cancel a booked flight, edit a booked flight, view all booked flights, add a bank account, track flight status, edit profile, and report/give feedback. The application is linked to the airline's office database to view customers the available flights and save their data, as well as to bank databases to provide the ability to pay booking costs and get refunds if a flight is cancelled.

1.3 Technologies Used

1 – Java programming language

2 – Android studio for mobile application development

3 – MYSQL Workbench for databases

1.4 Intended Audience

The audience for this SRS is the customer who want to make use of the application's service. The developer who must implement a high performance and secure application that suits the user requirements. The software tester who must develop valid test cases for the application until it reaches the acceptance of the customer. The project manager, who will utilize the SRS to carry out his duties, which include creating a detailed schedule to finish the project in the allotted time.

1.5 Overview

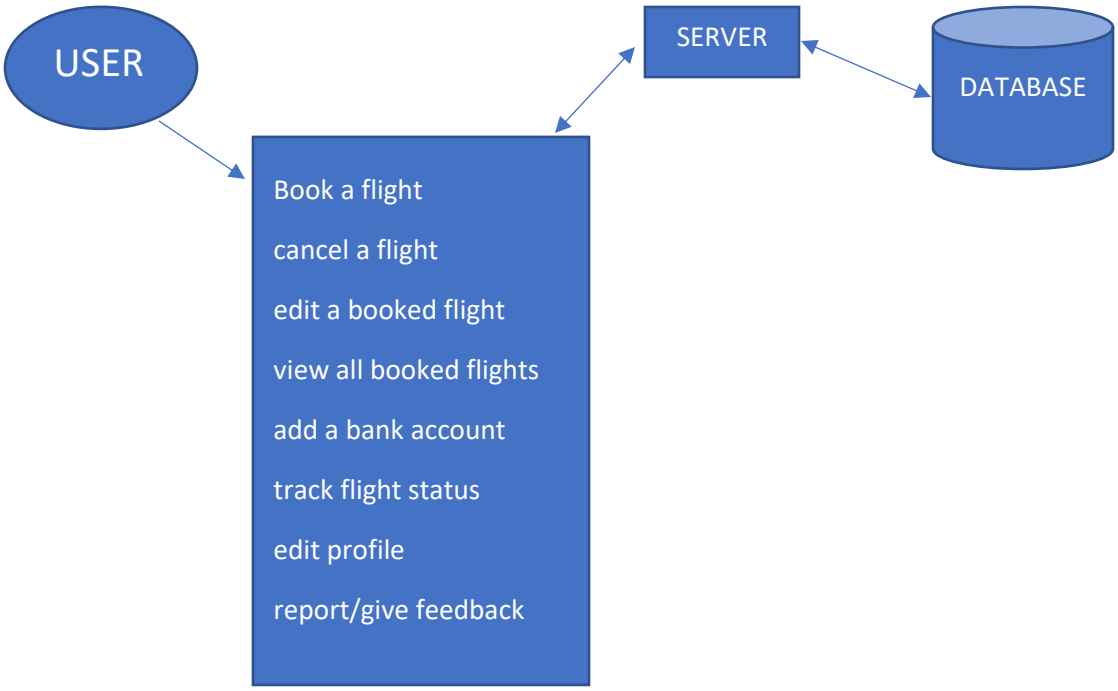
The SRS's remaining sections include explanations of the "Skies" Application's precise requirements. Chapter.2 lists the broad elements, like project restrictions and user characteristics, that have an impact on the Application and its requirements. Chapter.3 illustrates the details of

interfaces, and then the rest of the chapters' present the functional and nonfunctional requirements, Diagrams, and other related requirements of the application.

2. Overall Description

2.1 Product Perspective

Since "Skies" is an android-based application, an android device running on version 6.0 or higher is required. The app has a suitable user interface and is a self-contained, independent system. It will be able to establish a connection with a remote database server for the purposes of data storage, retrieval, and searching related to Clients, Flight information, and Reservation details. Clients are the only type of users who can use the system with certain functionalities.



Simple Block Diagram for "Skies" Android Application

2.2 User Characteristics

The system has one types of users: regular users (Clients) who have registered with the system and want to make use of the service without engaging in the process physically in an airline office by booking / canceling / editing a booked flight and many other

functionalities that are available through the interaction with the mobile application and will be discussed later in detail in the following sections.

2.3 Operating environment

The software being developed on Windows OS with the Android SDK using Android Studio will be deployed and run-on Android OS version 6.0 or above.

2.4 Constraints

- Android must be the operating system used by the application.
- The device of the user must provide network connectivity (WIFI).
- User's must have an account on the application.
- Java and the Android SDK are for software development.
- SQL server database engine is used by the system.

2.5 Assumptions and dependencies

It is assumed that the users of the application have the following:

- Android version compatible with the application.
- Internet connection.
- An account on the application linked with a banking account.

The system is assumed to depend on a reliable database for the for the purposes of data storage, retrieval, and searching related to Clients, Flight information, and Reservation details.

3.Interfaces

3.1 System interface

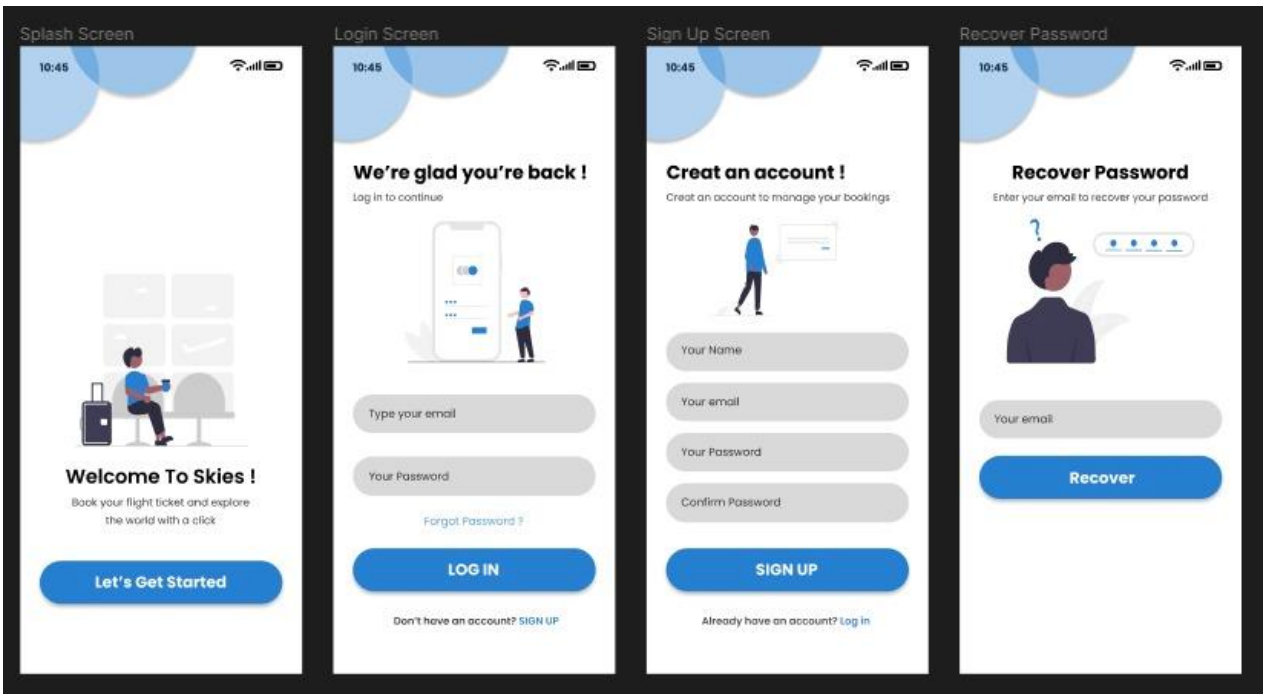
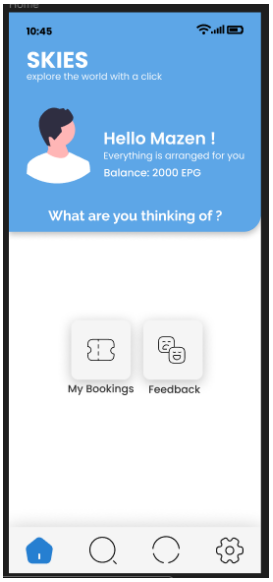


Figure.1

After the user launches the application, he/she will be able to login and start using the functionality of the if they already have been registered or they can register and account in case if they don't have one as shown in **Figure.1.** The registered user can recover the password of the account in case of forgetting it.



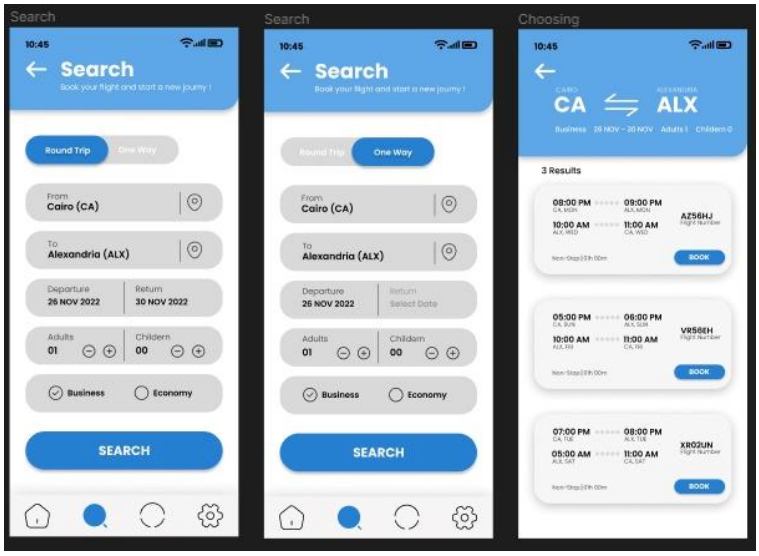


Figure.2 **Figure.3**

After the user successfully login/register to the application he/she will be directed to the home page as in **Figure.2** and be able to search for a flight as in **Figure.3**, the clicks on Book to book the desired flight after getting the results of the search.

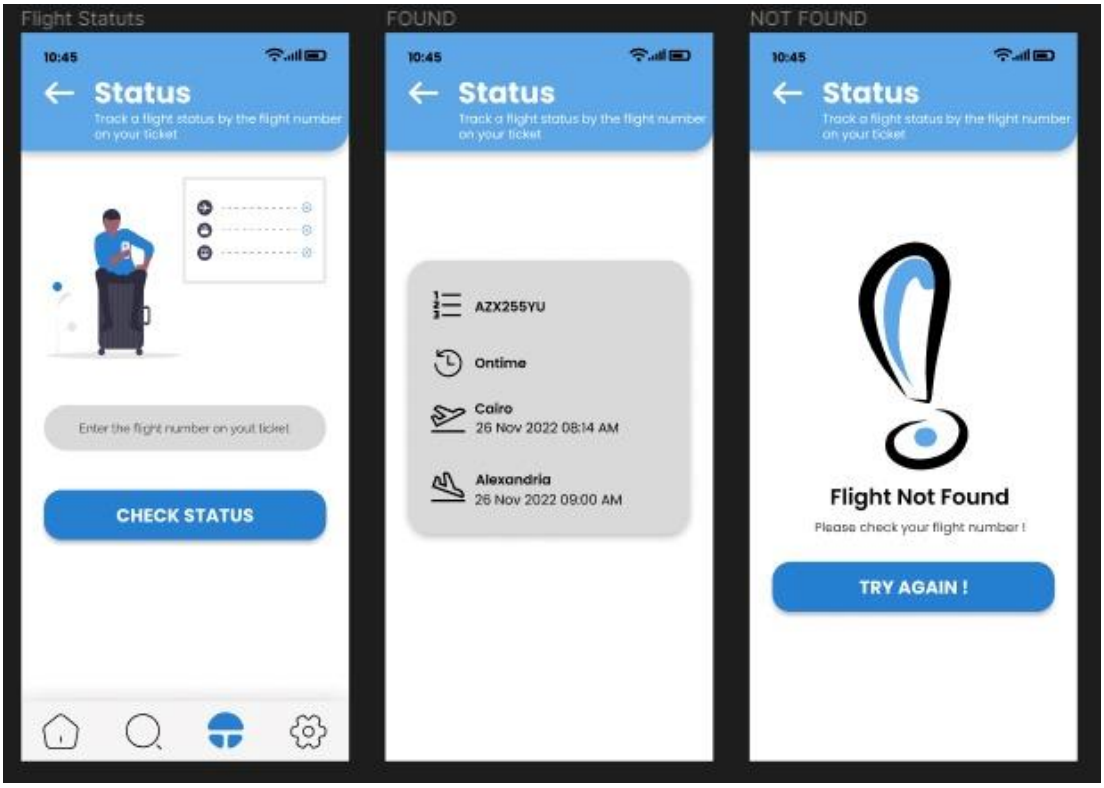


Figure.4

As in **Figure.4** the user can track a flight status by the flight number even the didn't book that flight, in case of providing an invalid flight number a massage will be shown to the user to check the number of the flight and try again.

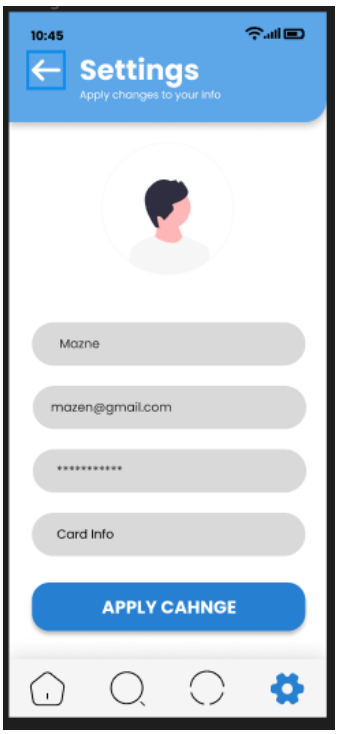


Figure.5

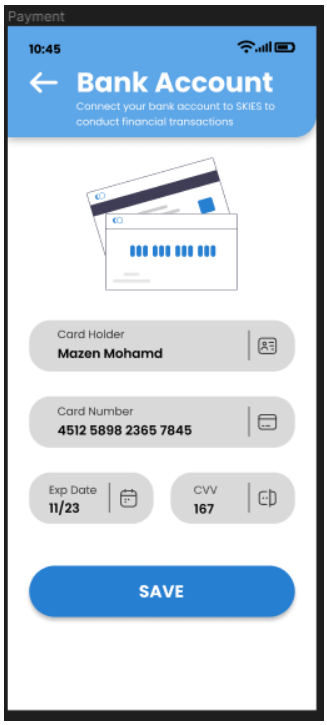


Figure.6

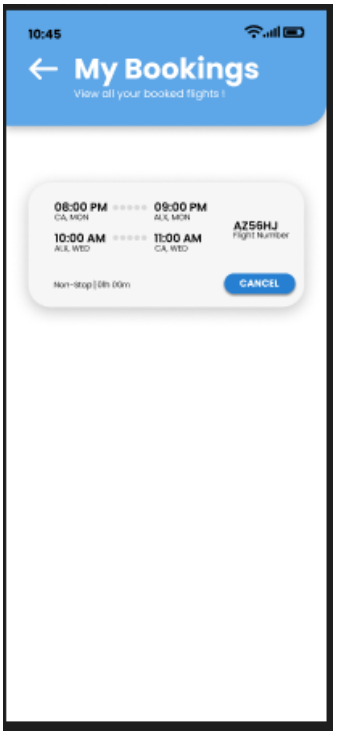


Figure.7

The user can edit the profile information such as name, email, etc. as shown in **Figure.5**. Also, as in **Figure.6** the user can add his/her account to manage financial transactions. In **Figure.7** the user can show all flights booked by him/her and cancel in case they want to do so.

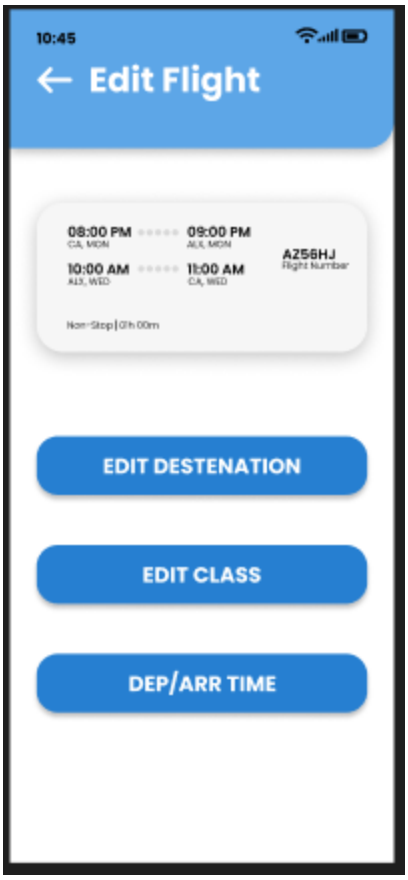


Figure.8

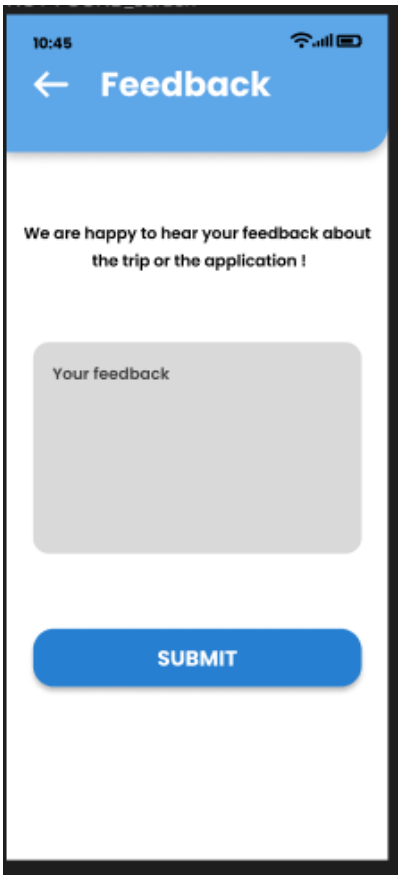


Figure.9

In **Figure.8** the user can edit a booked ticket and choose from the available option what they would like to edit and In **Figure.9** the user is able to give feedback about the trip or the service of the application.

3.2 Software Interface

Skies is an Airline reservations system app that is a native mobile application that will be developed under the Android operating systems environment using the Java JDK and the Android SDK tools. Android Studio, packages and libraries are used. System runs on android version 6.0 and higher versions. System interacts with database on the server side. System shall communicate with <mysql workbench> to run <mysql> data queries. System acts as a mediator between the airline and the client to make air travel reservations without the client having to visit the airline's office by receiving some data from the user like information or some transaction.

4. Functional requirements

4.1 Functional requirements

Title: User Login.

Description: System shall allow the registered users to login by providing a valid email and valid password.

4.2 Functional requirements

Title: User Signup.

Description: System shall allow non-registered users to register by providing a name, email, and password.

4.3 Functional requirements

Title: Book Flight.

Description: System shall allow user to enter his preferred booking schedule by choosing ticket types and processing online payment.

4.4 Functional requirements

Title: Cancel Booking.

Description: System shall allow user can request the cancellation of the booked ticket with 60% refund by maximum a week.

4.6 Functional requirements

Title: View Booking.

Description: System shall display all his previous booking data.

4.7 Functional requirements

Title: Link bank account.

Description: System shall prompt the user to their banking account with it.

4.8 Functional requirements

Title: Track Flight Status.

Description: System shall display the flight statues by its number to show whether it is cancelled, changed flight time or still on time.

4.9 Functional requirements

Title: Edit Profile.

Description: System shall allow user to change his information (user data, name, bank account).

4.10 Functional requirements

Title: Report/Give Feedback.

Description: System shall display a text filed for the user to give feedback.

4.11 Functional requirements

Title: Forget Password

Description: System shall display the users' password in case of forgetting it by providing the users' email.

4.12 Functional requirements

Title: Edit ticket details

Description: System shall allow user to edit ticket details such as destination, class, and arrival / departure time.

5. Non-Functional requirements

5.1 Security Requirements.

- Customized password should be at least 8 characters.

5.2 Availability Requirements.

- The system should be available 24/7.

5.3 Portability Requirements.

- System should be manageable android 6.0 while being lightweight mobile application.

5.4 Safety Requirements.

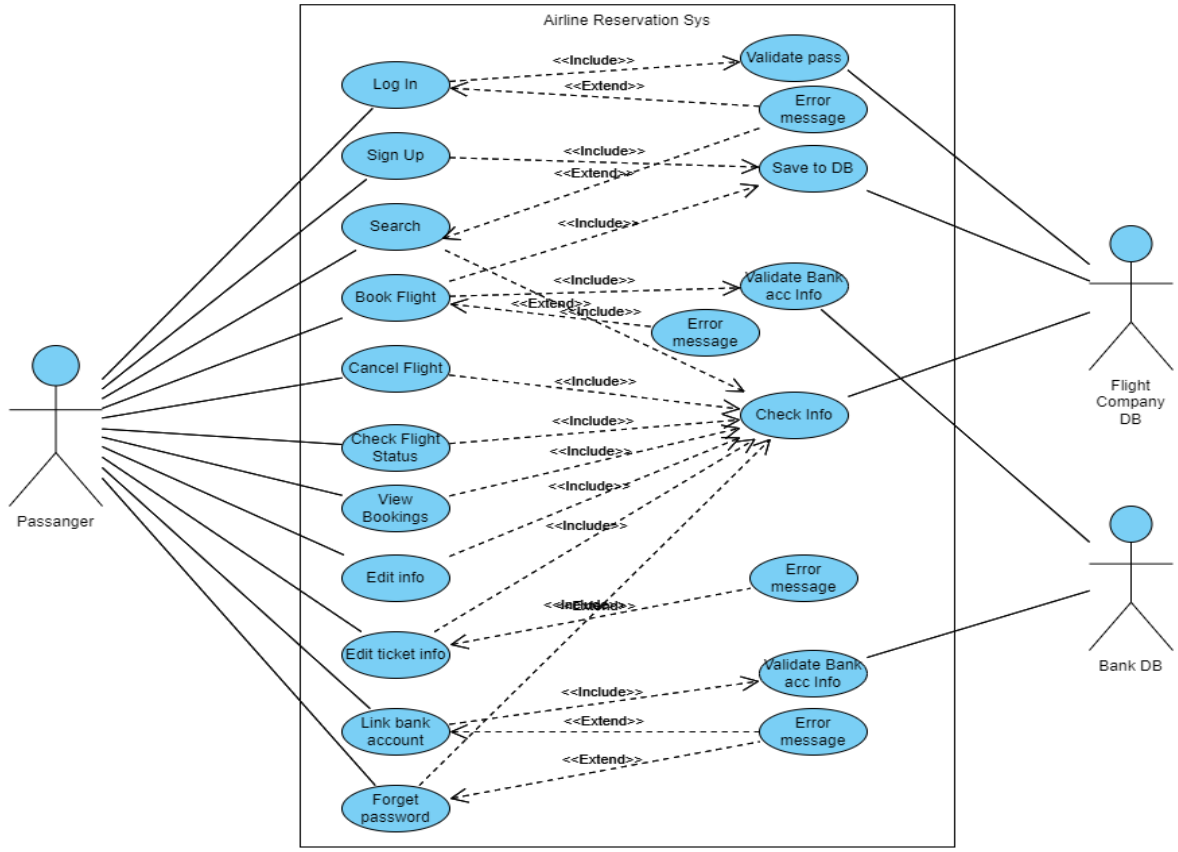
- System should assure data integrity.

5.5 Safety Requirements.

- System should include restore and recover functions to prevent data loss.

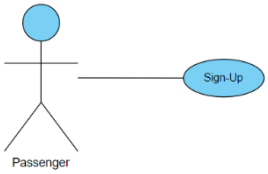
4. Diagrams

4.1 Use case diagram



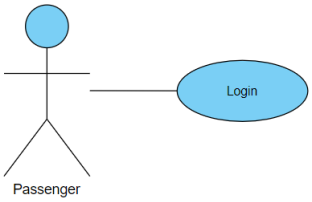
4.2 Use case scenarios

Use case 01



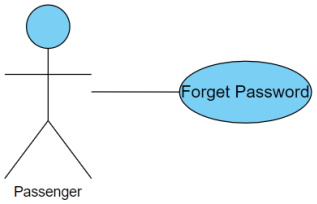
Usecase Name	***Sign-Up***
Actors	***Passenger (Non-registered).***
Main success scenario:	1. ***The system asks for an e-mail address and password, or the user can register for a new account.*** 2. ***The user selects registration.*** 3. ***The System prompts user for registration information, Name, E-mail, password, Confirm password.*** 4. ***The user provides the required information.*** 5. ***User clicks on submit button (Sign Up)*** 6. ***System checks if the username is not already in use.*** 7. ***System verifies information and creates account.***
Exceptions	***4a. User clicks submit without entering the required information that the system asked for.*** ***6a. The E-mail already exists in the system.***
Actions	***4.1 System display error alert to the user "Information required".*** ***4.2 User refill required info.*** ***6.1 System asks the user to Login "Username already have an account".***
Pre-condition	***- User has registered successfully.***
Post condition	***- Number of users of the system is increased by 1*** ***- User is successfully registered*** ***- Login info is stored in the DB***

Use case 02

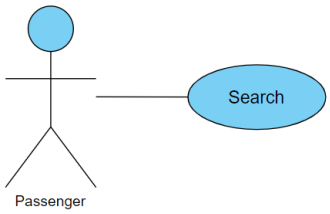


Usecase Name	***Login***
Actors	***Passenger.***
Main success scenario:	1. ***System displays login page for User (Passenger) and requires E-mail and password.*** 2. ***User provides the E-mail and valid password.*** 3. ***User clicks on submit button <Login>.*** 4. ***System validates the User E-mail and password and logs him/her into the system.***
Exceptions	2a User typed invalid E-mail or invalid password. 3a User tried to login with an empty field.
Actions	***2.1 System displays an error message like "Invalid E-mail OR Password, Please Try again".*** ***2.2 User retries to login OR cancels operation.*** ***3.1 System displays an error message like "please enter your E-mail/password".*** ***3.2User retries to login OR cancels operation.***
Pre-condition	***User has an account.***
Post condition	1. ***The User (Passenger) is logged in to the system.*** 2. ***The User has access to the functions (For Passenger) of the system.***

Use case 03

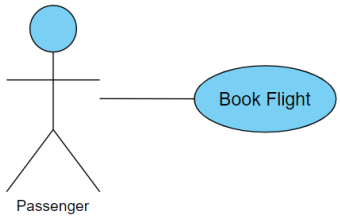
	
Usecase Name	***Forget Password***
Actors	***Passenger.***
Main success scenario:	***1. System displays Recover password page for User (Passenger) and requires E-mail.*** ***2. User provides valid E-mail.*** ***3. User clicks on submit button < Recover >.*** ***4. System validates the user's E-mail address and sends the password to him/her via E-mail.***
Exceptions	2a User typed invalid E-mail. 3a User tried to Recover with an empty field.
Actions	***2.1 System displays an error message like "Invalid E-mail, please enter the correct e-mail".*** ***2.2User retries to Recover OR cancels operation.*** ***3.1 System displays an error message like "please enter your E-mail ".*** ***3.2 User retries to Recover OR cancels operation.***
Pre-condition	***1. User has an account.*** ***2. User cannot logged in.***
Post condition	***1. The User Recovering him/her password.*** ***2. The User (Passenger) is logged in to the system.*** ***3. The User has access to the functions (For Passenger) of the system.***

Use case 04



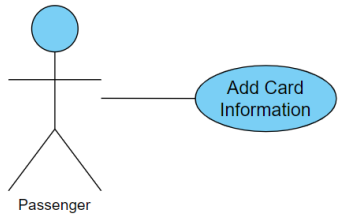
Usecase Name	***Search***
Actors	***Passenger.***
Main success scenario:	1. ***The System prompts user for trip details such as the Departure (From), Destination (To), Departure Date, Return Date (if Rounded Trip), etc.*** 2. ***The user provides the required detail.*** 3. ***User clicks on submit button <SEARCH>.*** 4. ***The system checks the available flight schedules.*** 5. ***System display results.***
Exceptions	4a User tried to Search with an empty field. 5a User searches for flights that are not currently available.
Actions	***4.1 System displays an error message like "please complete flight details ".*** ***4.2 User retries to Research OR cancels operation.*** ***5.1 System displays an error message like "your search is currently unavailable ".*** ***5.2 User search in different details OR cancels operation.***
Pre-condition	***User logged in.***
Post condition	1. ***The user has access to all the results corresponding to his/her search.*** 2. ***User can access all flights details.***

Use case 05



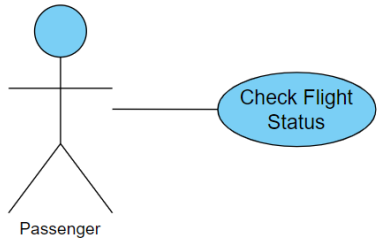
Usecase Name	***Book Flight***
Actors	***Passenger.***
Main success scenario:	1. ***User choose the flight wants to book.*** 2. ***User clicks on book button <BOOK>.*** 3. ***System validates the user's card balance.*** 4. ***System verifies(confirm) card balance.*** 5. ***System withdrawals funds of the flight from the user's account and books the flight.***
Exceptions	2a The user has not added a credit card. 4a User does not have enough balance.
Actions	***2.1 the system will display Bank Account page add card information then check balance.*** ***2.2 User add card information OR cancels operation.*** ***4.1 System displays an error message like "Please, add enough balance too book your flight ".***
Pre-condition	***-User logged in*** ***-User have an account..***
Post condition	***User booked his/her flight successfully.***

Use case 06



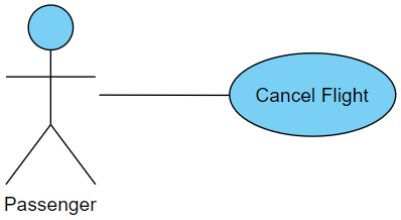
Usecase Name	***Add Card information***
Actors	***Passenger.***
Main success scenario:	1. ***System displays Bank Account page for User (Passenger) to add card information and requires Card holder, Card number, Exp date, Card verification value (CVV).*** 2. ***User provides the required information.*** 3. ***User clicks on submit button <SAVE>.*** 4. ***System validates the Card information and saves in DB.***
Exceptions	2a User type invalid information like CVV, etc. 3a User save without data or data missing
Actions	***2.1 System display error message like "Please, enter valid CVV, etc."*** ***2.2 User retries enter the correct data or cancel operation*** ***3.1 System display error message like "Please, enter valid Card number, etc."*** ***3.2 User retries to add the missing information or cancel operation***
Pre-condition	***User logged in.***
Post condition	1. ***The User (Passenger) added credit card to the system.*** 2. ***The User has access to book flight.***

Use case 08

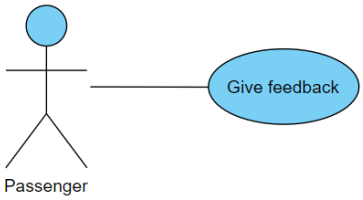


Usecase Name	***Check Flight Status***
Actors	***Passenger.***
Main success scenario:	1. ***User clicks on status page icon to open.*** 2. ***System displays status page for User and requires flight number in his/her ticket.*** 3. ***User clicks on submit button <CHECK STATUS>.*** 4. ***System validates the User flight number.*** 5. ***System switch to another status page and display flight information.***
Exceptions	2a User typed invalid flight number. 3a User tried to check flight status with an empty field.
Actions	***2.1 System display flight not fount page and display error message like "please, check your flight number !" and asks user to try again.*** ***2.2 User retries to enter correct flight number OR cancels operation.*** ***3.1 System display error message like "please, enter your flight number"*** ***3.2 User retries enters a flight number OR cancels operation.***
Pre-condition	***User logged in.***
Post condition	***User view his/her flight status.***

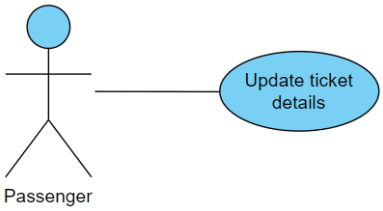
Use case 09

Sign-Up 	
Usecase Name	***Cancel Flight***
Actors	***Passenger.***
Main success scenario:	**1.** ***User enters my booking page.*** **2.** ***System display all booked flights by the user.*** **3.** ***User chooses the flight he wants to cancel.*** **4.** ***User clicks on the cancel flight button <CANCEL>.*** **5.** ***System cancels the trip and updates my booking page and DB.***
Pre-condition	***User logged in.***
Post condition	***Flight is canceled in my booking page.***

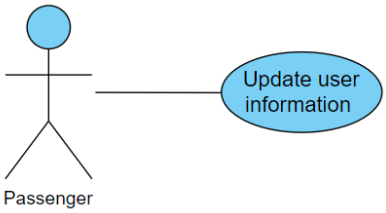
Use case 10

Sign-Up	
Usecase Name	***Give feedback***
Actors	***Passenger.***
Main success scenario:	**1.** ***User enters Give feedback page.*** **2.** ***System display textbook.*** **3.** ***User write her/his feedback.*** **4.** ***User clicks on the submit button <submit>.*** **5.** ***System saves the feedback in DB.***
Pre-condition	***User logged in.***
Post condition	***User add her/his feedback.***

Use case 11

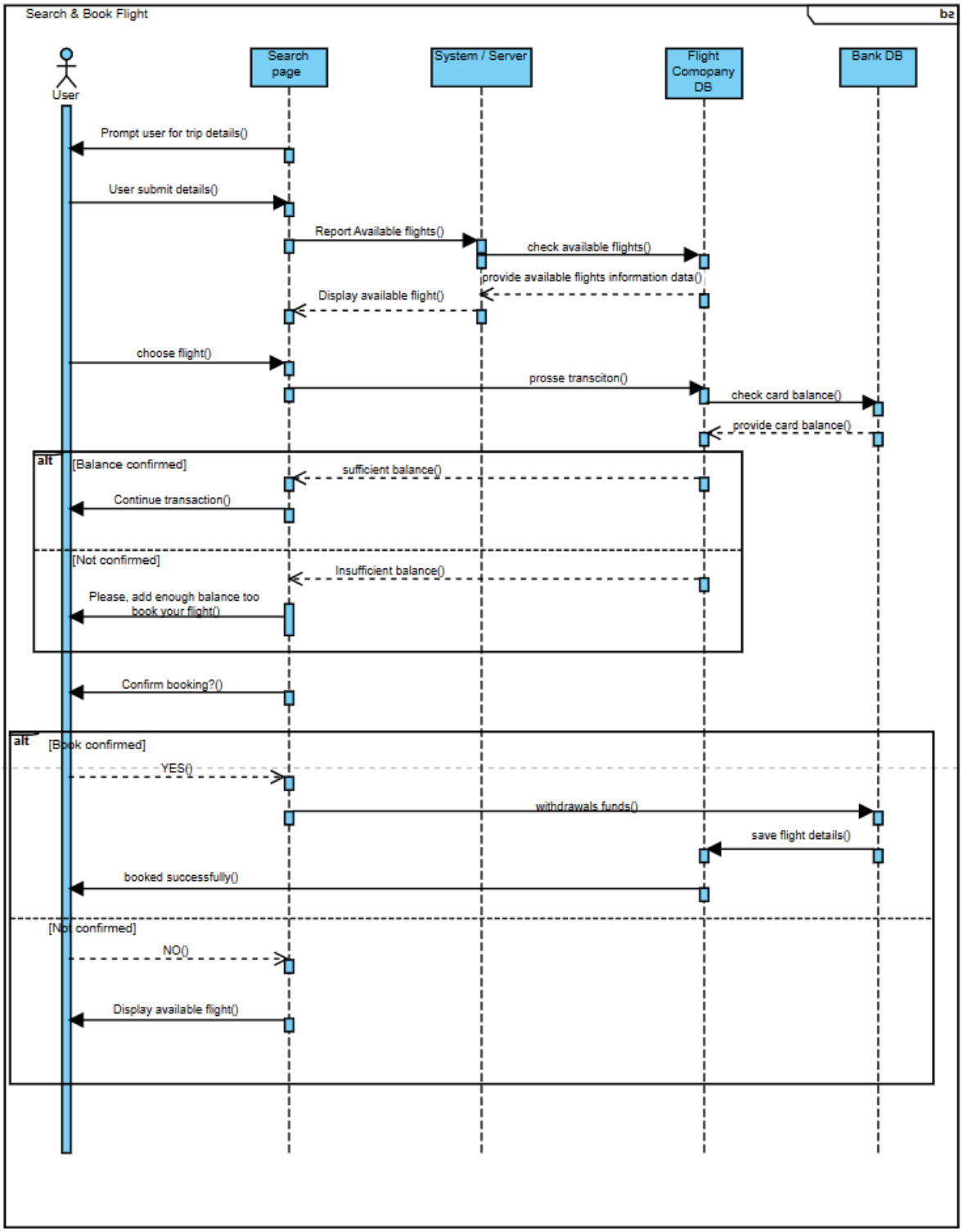
Sign-Up 	
Usecase Name	***Check Flight Status***
Actors	***Passenger.***
Main success scenario:	**1.** ***User enters update tickets details page.*** **2.** ***System displays details that the user can update like destination, class, etc.*** **3.** ***User chooses what he wants to update by clicking on it.*** **4.** ***User submits her/his changes.*** **5.** ***System checks the changes.*** **6.** ***System updates the changes in DB.***
Exceptions	5a User update or change for flights that are not currently available or for class fully booked.
Actions	***5.1System displays an error message like "Sorry you can't change "class to economy" as "its fully booked", please check the flight status ".*** ***5.2 User cancels operation OR cancel flight and book another flight.***
Pre-condition	***User logged in.***
Post condition	**Old flight details is updated to the new one**

Use case 12

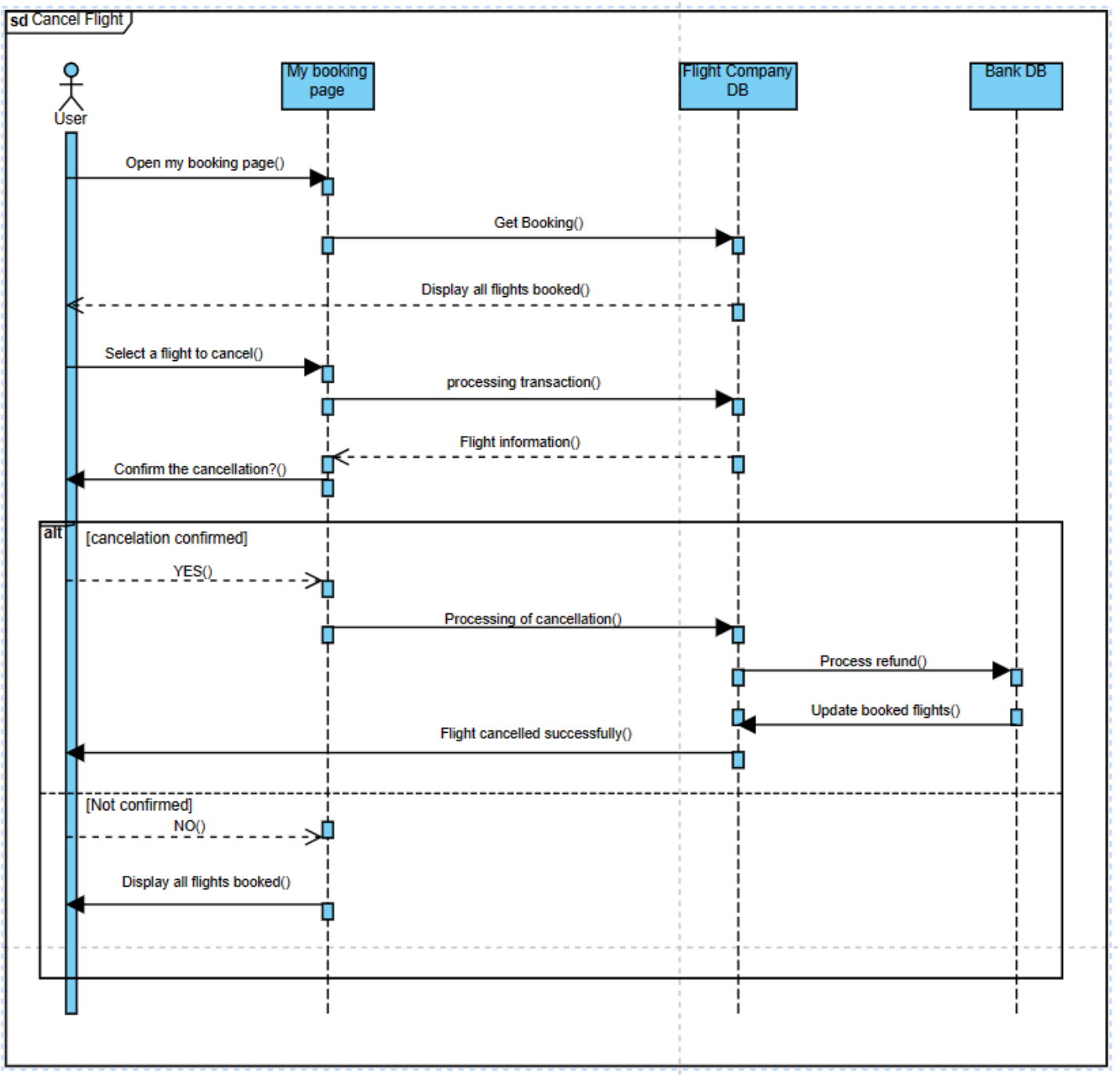
Sign-Up 	
Usecase Name	***Update user information***
Actors	***Passenger.***
Main success scenario:	**1.** ***User enters update info page.*** **2.** ***System displays user information like Name, E-mail, etc.*** **3.** ***User adds information he wants to update.*** **4.** ***User submits her/his changes by clicking on submit button <APPLY CHANGE>.*** **5.** ***System checks the changes.*** **6.** ***System verifies and updates the information in DB.***
Exceptions	5a User updates her/his info to invalid information or already exists.
Actions	***5.1 System displays an error message like "Invalid E-mail OR E-mail already exists, Please Try again".*** ***5.2User retries to update OR cancels operation.***
Pre-condition	***User logged in.***
Post condition	**Old user information is updated to the new one**

6.3 Sequence diagram

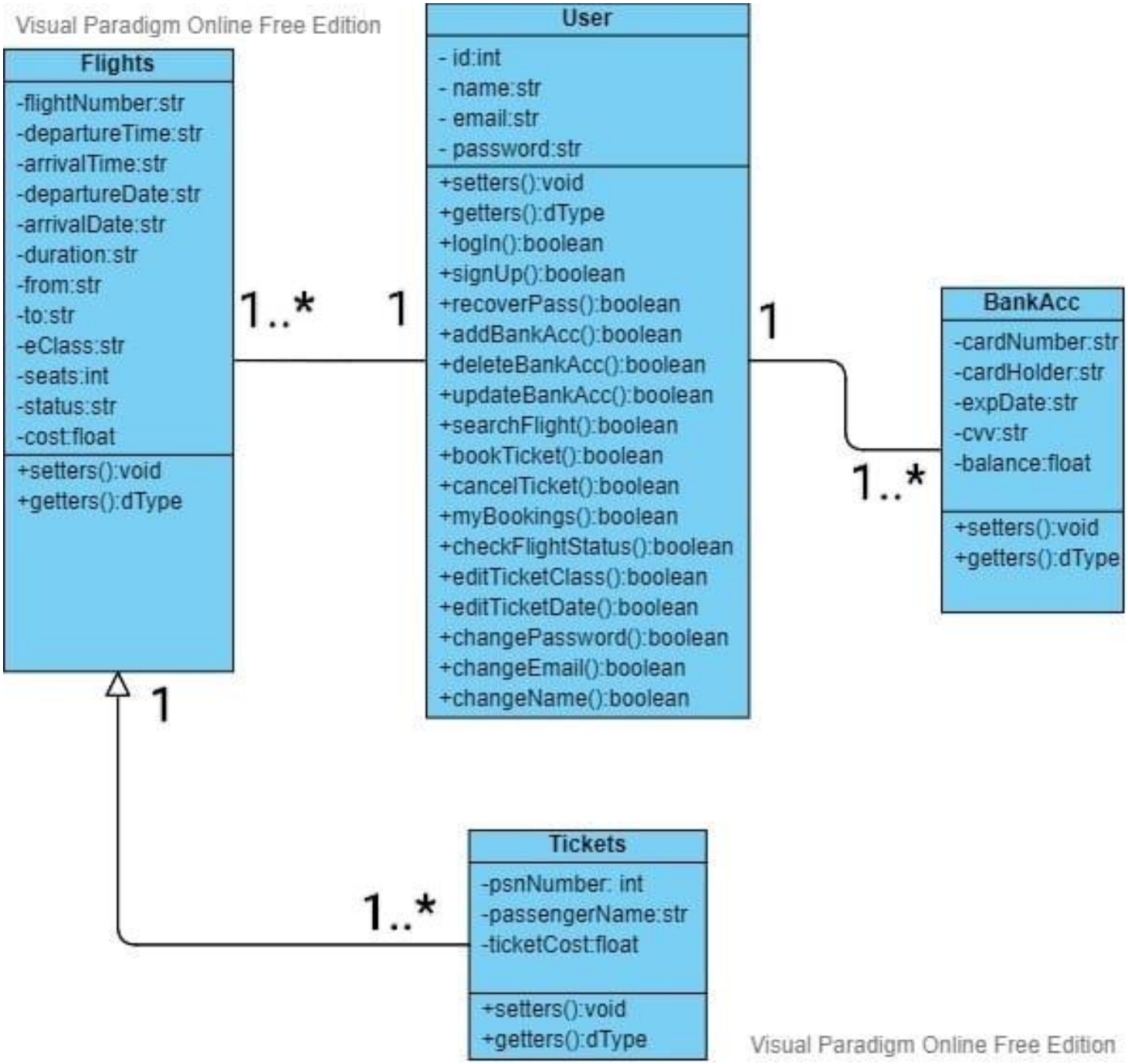
6.3.1 Search & Book Flight



6.3.2 cancel Flight



UML Class diagram



Testing

7.1 Test Cases

Project Name:

Test Case 1

Test Case ID: TU01

Test Priority (Low/Medium/High): Med

Module Name: ***Sign-Up screen***

Test Title: Verify sign-up with valid E-mail and password

Description: Test the sign-up functionality in App

Test Designed by: Mazen

Test Designed date: 2022\12\20

Test Executed by: Mazen

Test Execution date: 2022\12\21

Pre-conditions: User has valid E-mail and password

Dependencies:

Step	Test Steps	Test Data	Expected Result	Actual Result	Status (Pass/Fail)	Notes
1	Navigate to Sign-up page	Sign-up button	Page should open	As Expected	pass	
2	Provide password	P@1234	Credential can be entered	As Expected	pass	
3	Enter valid E-mail in invalid format	gmail.com @Ahmad	“Invalid format” message should be display	It accept invalid format	Fail	
4	Clicks on submit button <Sign Up>		User is registered	As Expected	pass	

Post-conditions:
Validates user data and saves data in the database.

Project Name:

Test Case 2

Test Case ID: TU01

Test Priority (Low/Medium/High): Med

Module Name: login screen

Test Title: Verify login with valid username and password

Description: Check Customer Login with valid Data

Test Designed by: Mazen

Test Designed date: 2022\12\20

Test Executed by: john Anwar

Test Execution date: 2022\12\21

Pre-conditions: User has valid username and password

Dependencies:

Step	Test Steps	Test Data	Expected Result	Actual Result	Status (Pass/Fail)	Notes
1	Go to app	Username= guru99	User should Login into an application	As Expected	Pass	
2	Enter Username	Password: pass99				
3	Enter Password					

4	Click Submit					

Post-conditions:
User is validated with database and successfully login to account. The account session details are logged in database.

Project Name:

Test Case 3

Test Case ID: TU02

Test Priority (Low/Medium/High): High

Module Name: Booking

Test Title: Book Flight

Description: Ensure That flights can be booked using the app

Test Designed by: Mazen

Test Designed date: 2022\12\20

Test Executed by: SEIF ELDIN SAMIR

Test Execution date: 2022\12\22

Pre-conditions: User is logged in successfully and the flight is already available

Dependencies:

Step	Test Steps	Test Data	Expected Result	Actual Result	Status (Pass/Fail)	Notes
1	Go to Search the section				Pass	
2	Enter Date of Flight:	Date:1/1/2023				
3	Enter Where you are Flying from:	From: Egypt				

4	Enter Where you are Flying To:	To: France				
5	Choose A Flight		Flight dialog Appears	As expected The dialog appears		
6	Click the Ok Button					
7	Enter Your Name	SEIF ELDIN SAMIR				
8	Procced to Payment		After payment number of seats decreases by 1	Number of seats decreased accordingly		

Project Name:

Test Case 4

Test Case ID: TU01

Test Priority (Low/Medium/High): Med

Module Name: ***Cancel Flight***

Test Title: cancel booked flight

Test Designed by: Mazen

Test Designed date: 2022\12\20

Test Executed by: Ahmad Mustafa

Test Execution date: 2022\12\21

Description: Test the cancel flight functionality in App

Pre-conditions: User has valid account

Dependencies:

Step	Test Steps	Test Data	Expected Result	Actual Result	Status (Pass/Fail)	Notes
1	Navigate to my booking page	My Booking button	Page should open	As Expected	pass	
2	Clicks on cancel button <CANCLE>		Flight was cancelled	As Expected	pass	
3	Cancelling a flight and receiving a refund	Clicks on cancel button	receiving a refund	As Expected	pass	

Post-conditions:
Cancelling the flight and, Update database

Project Name:

Test Case 4

Test Case ID: TU01

Test Priority (Low/Medium/High): Med

Module Name: ***Cancel Flight***

Test Title: cancel booked flight

Test Designed by: Mazen

Test Designed date: 2022\12\20

Test Executed by: Ahmad Mustafa

Test Execution date: 2022\12\21

Description: Test the cancel flight functionality in App

Pre-conditions: User has valid account

Dependencies:

Step	Test Steps	Test Data	Expected Result	Actual Result	Status (Pass/Fail)	Notes
1	Navigate to my booking page	My Booking button	Page should open	As Expected	pass	
2	Clicks on cancel button <CANCLE>		Flight was cancelled	As Expected	pass	
3	Cancelling a flight and receiving a refund	Clicks on cancel button	receiving a refund	As Expected	pass	

Post-conditions:
Cancelling the flight and, Update database

Project Name:

Test Case 5

Test Case ID: TU01

Test Priority (Low/Medium/High): Med

Module Name: Check Flight Statues

Test Title: Check the latest updates about the booked flight

Description: Check the booked details and the remaining time for the flight

Test Designed by: Mazen

Test Designed date: 2022\12\20

Test Executed by: Mazen Sarwat

Test Execution date: 2022\12\21

Pre-conditions: there Customer has booked the flight or searched the flight

Dependencies: Known flight number

Step	Test Steps	Test Data	Expected Result	Actual Result	Status (Pass/Fail)	Notes
1	Enter the flight number for verification	Flight no.: AZX255YU	Flight statues should appear. Flight details is shown.	As Expected	pass	
2	Press check statues button		Arrival and departure time is shown	As Expected	pass	

Post-conditions:
flight number is validated with database and successfully login to account.